Princeton University

ORF 363/COS 323
Final Exam, Fall 2022

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1. Please write your name on the first page of your solutions. Next to it, please write out and sign the following pledge: "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this examination."
2. The exam is not to be discussed with *anyone* except possibly the instructors and the AIs. You can only ask *clarification questions*, and only as *public* (and preferably non-anonymous) questions on Ed Discussion. No emails.
3. You are allowed to consult the lecture notes, your own notes, the reference books of the course as indicated on the syllabus, the problem sets and their solutions (yours and ours), the midterm and its solutions (yours and ours), the practice midterm and final exams and their solutions, all Ed Discussion posts, but *nothing else*. You can only use the Internet in case you run into problems related to software.
4. You may refer to facts proven in the notes or problem sets without reproving them.
5. For all computational problems, include your code. The output that you present should come from your code.
6. Unless you have been granted an extension because of overlapping finals, the exam is to be turned in on Tuesday (December 20, 2022) at 10 AM EST on Gradescope as a single PDF file. You are free to write your solutions on paper or on a tablet, or to type them up. Only the latest version submitted before your deadline will be graded.
7. Each question has 20 points. You need to justify your answers to receive full credit.

Problem 1: Convexity

(a) Show that $x = (0, 0, 0)$ is a strict global minimum of the function:

$$f(x, y, z) = x^2 + y^2 + z^2 - xy + yz - xz. \tag{1}$$

(b) Consider the function

$$f(x, y) = e^{x-y} + e^{y-x}. \tag{2}$$

Show that all the points $(x, y) \in \mathbb{R}^2$ with $x = y$ are global minimizers of f (or equivalently, all the points on the line $x - y = 0$ are global minimizers).

(c) Let f be a function from $\mathbb{R}^n$ to $\mathbb{R}$. Recall an affine function is one that can be written as $f(x) = a^T x + b$ for some vector a and some scalar b. For the following statements, provide a proof or a counterexample.

(i) If f is both convex and concave, then f is affine.
Hint: You may assume f is twice differentiable.

(ii) If $g \circ f$ is convex for all convex functions g, then f is affine.

Problem 2: Nonlinear least squares

On a recent camping trip to Africa, our friend Alice was fascinated with the beauty of two flowers (let's call them A and B) that were growing close to her tent. She decided to collect measurements of their growth and use the knowledge she acquired during her college years to develop a mathematical model that fits the data well and then use it to predict the growth of the flowers. She collected measurement during certain days only, because she was busy exploring several villages and towns near the camp she was staying. The measurements for flowers A and B are as follows:

$$A = \{(1, 3.2939), (2, 4.2699), (4, 7.1749), (5, 9.3008), (8, 20.259), (10, 36.12)\},$$
$$B = \{(1, 3.023), (2, 4.36), (4, 6.258), (5, 8.21), (8, 17.85), (10, 21.25)\}.$$

The elements in the above datasets have the following structure (t, y), where t is the day Alice measured the flowers and y is the actual measurement.

Alice's friend John told her that she can try to fit an exponential growth model of the form

$$T(t) = c_1 e^{c_2 t}. \tag{3}$$

Alice recalled that she can use nonlinear least squares to determine the unknown parameters c_1 and c_2 in the model.

(a) Write the nonlinear least squares problem that minimizes the sum of the squares of the residuals between the data and the prediction from the growth model.

(b) Apply 15 iterations of the Gauss-Newton method to determine two separate models, one for flower A and another for flower B. For both models use the same starting point $(c_1^0, c_2^0) = (2.75, 0.275)$. Plot Alice's measurements (ground truth) and the best fit that you found on the same figure (one plot for plant A and another for plant B).

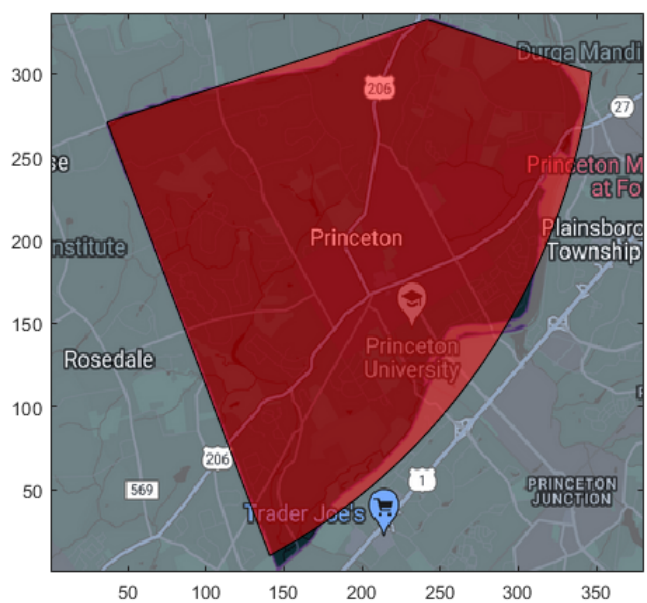
(c) Discuss the relation between the objective function value of nonlinear least squares and how well the resulting model fits the data.

Problem 3: Radio Free Princeton

The town of Princeton is constructing an antenna to improve cell coverage. You have been hired to decide the antenna's location and range. The signal will reach all points whose euclidean distance to the antenna's location is less than the antenna's range. Due to possible interference with other towns' cell networks, you must ensure the signal only reaches points within the town's territory. Subject to this constraint, you want the signal to reach as large an area as possible. The town's territory can be described by the following convex set:

$$P = \left\{ x \in \mathbb{R}^2 \quad \text{s.t.} \quad \begin{array}{c} x_1 + 0.4x_2 \geq 145 \\ 0.3x_1 - x_2 >= -260 \\ 0.3x_1 + x_2 <= 405 \\ ||x - (-30, 350)|| <= 380 \end{array} \right\}.$$

The set is an intersection of three half-spaces (the first three constraints) and a circle (the last constraint.) As you can see below, this is actually just an approximation, but for the purposes of this problem, you can assume this is an exact description.



(a) First you have to do some theoretical work. Consider the circle $\mathcal{B}$ with center x_b and radius r_b defined as

$$\mathcal{B} = \{x \in \mathbb{R}^2 \mid ||x - x_b|| \leq r_b\}.$$

(i) Consider the half-space $\mathcal{H} = \left\{x \in \mathbb{R}^2 | a^T x \leq b\right\}$. Show that the constraint $\mathcal{B} \subseteq \mathcal{H}$ can be written (or expressed) as follows:

$$a^T x_b + r_b ||a|| \leq b.$$

Hint: $\mathcal{B} \subseteq \mathcal{H}$ is equivalent to $\max_{x \in \mathcal{B}} a^T x \leq b$

(ii) Consider another circle $\mathcal{C} = \{x \in \mathbb{R}^2 \mid ||x - x_c|| \leq r_c\}$. Show that the constraint $\mathcal{B} \subseteq \mathcal{C}$ can be written (or expressed) as follows:

$$||x_b - x_c|| + r_b \leq r_c.$$

Hint: $\mathcal{B} \subseteq \mathcal{C}$ is equivalent to $\max_{x \in \mathcal{B}} ||x - x_c|| \leq r_c$.

(b) Show that the optimal location and range can be found with a convex problem.

(c) Solve the convex problem using CVX and report the optimal location and range.

Problem 4: Miracle on 363rd Street

This holiday season, you are feeling generous and want to give away presents to the children in your neighborhood. There are n children so you have bought n presents. You will give one present to each child. All that is left is to decide which present to give to which child. Let E_{ij} be the enjoyment child i will receive from present j. You wish to maximize the total enjoyment of all the children.

(a) Formulate the problem of maximizing total enjoyment as an integer program.
Hint: Have a binary variable for each pair (i, j).

(b) Take the LP relaxation of your integer program.

(c) Using CVX, solve your derived LP with the matrix E provided in the file `Miracle.mat`.

(d) Report an optimal present distribution.

Problem 5: SDP

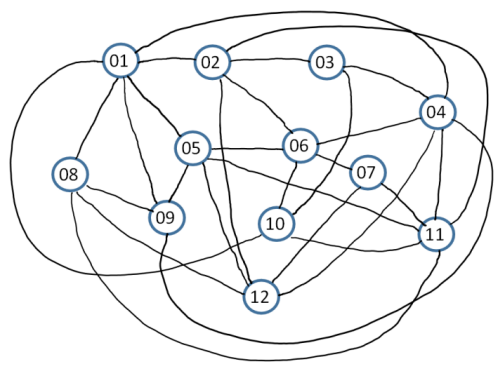
(a) **Conversion to SDP:** Consider the following optimization problem

$$\begin{aligned} \max_{x_1, x_2 \in \mathbb{R}} \quad & 3x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 10, \\ & x_1 x_2 \geq 4, \\ & x_1 + x_2 \geq 0. \end{aligned} \tag{4}$$

Write problem (4 as an SDP. Hint: consider the matrix

$$X = \begin{bmatrix} x_1 & 2 \\ 2 & x_2 \end{bmatrix}.$$

(b) **Let's finish where we started:** In the very first lecture, we tried to avoid "overlapping finals" by finding the largest stable set (aka independent set) in the following graph:



We found one of size 6, but at the time we couldn't show that 6 was the best possible without trying all possibilities. Solve a convex optimization problem (e.g., the LP relaxation) which proves that 6 is indeed optimal. Show your CVX code and its output. Argue precisely why the solution to your optimization problem certifies that 6 is the best possible. For your convenience, the adjacency matrix of the graph is provided below. You can copy and paste it into MATLAB.

$$G = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$